Biomedical importance of Carbohydrates

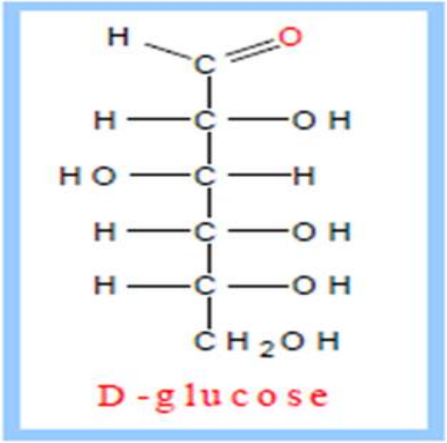
Course: BDS

Date: 24/06/16

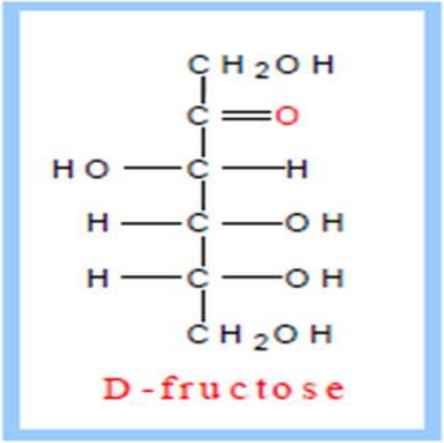
Define Carbohydrates?

DEFINITION

- **Carbohydrates are polyhydroxy aldehydes or ketones or compounds which yield these on hydrolysis.**



D-glucose



D-fructose

Mention the significance of carbohydrate?

BIOMEDICAL IMPORTANCE

1. Most abundant dietary source of energy.
2. Also serve as storage form of energy – Glycogen.
3. Participate in the structure of cell membrane & cellular functions (cell growth, adhesion and fertilization).
4. Mucopolysaccharides form the ground substance of mesenchymal tissues.
5. Certain carbohydrate derivatives are used as drugs, like cardiac glycosides / antibiotics.

Enlist carbohydrate related disorders

ASSOCIATED DISORDERS

- Derangement in Glucose metabolism – Diabetes Mellitus.
- Inherited deficiency of certain enzymes in metabolic pathways of different carbohydrates cause diseases.
  - Glycogen storage disorders
  - Galactosemia
  - Hereditary fructose intolerence
  - Lactose intolerance, etc.

Enlist the sources of carbohydrates



Classify carbohydrates

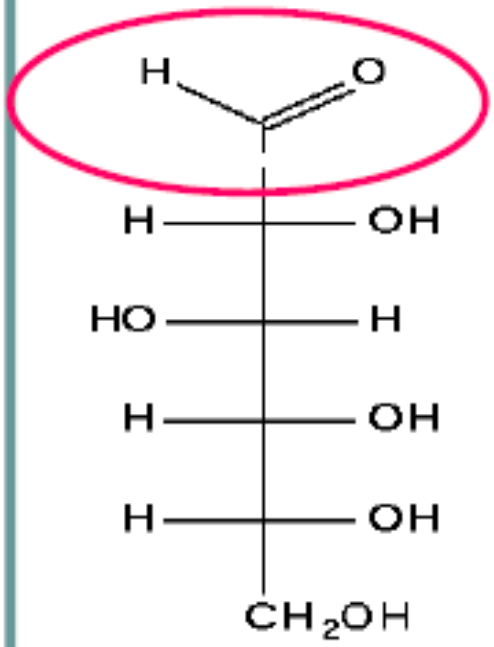
CLASSIFICATION

- Based on number of sugar units present.
  - **Monosaccharides**.
    - Cannot be hydrolyzed further into simpler forms.
  - **Disaccharides**.
    - Yield 2 molecules of same or different monosaccharide units on hydrolysis.
  - **Oligosaccharides**.
    - Yield 3-10 molecules of monosaccharide units on hydrolysis.
  - **Polysaccharides**.
    - Yield more than 10 molecules of same or different monosaccharide units on hydrolysis.
    - Homo- & Heteropolysaccharides.

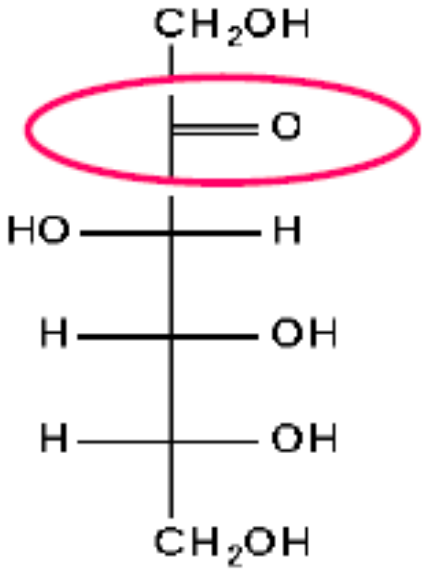
MONOSACCHARIDES

- Simplest group of carbohydrates, cannot be further hydrolysed.
- General formula : $C_n(H_2O)_n$
- Categorization of monosaccharides is based on
  - the Functional Group. (Aldehyde or keto)
  - the Number of Carbon atoms.

MONOSACCHARIDES BASED ON FUNCTIONAL GROUP



ALDOSE



KETOSE

Name the common carbohydrates

COMMON MONOSACCHARIDES

No. of C atoms	Generic name	Aldoses	Ketoses
3	Triose	Glyceraldehyde	Dihydroxy acetone
4	Tetrose	Erythrose	Erythrulose
5	Pentose	Ribose Xylose	Rilulose Xylulose
6	Hexose	Glucose Galactose	Fructose
7	Heptose	Glucoheptose	Sedoheptulose

GLYCOSIDE FORMATION

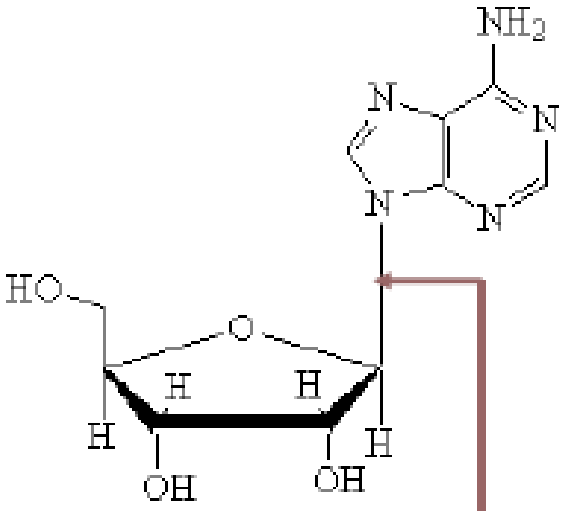
- The hydroxyl group of anomeric carbon of a carbohydrate can join with a hydroxyl group of another carbohydrate or some other compound to form a glycoside and the bond so formed is known as glycosidic bond.

  eg. R-OH + HO-R' → R-O-R' + H_2O

- The non-carbohydrate moiety is known as aglycone – phenol, sterol, bases, CH_3OH, glycerol.

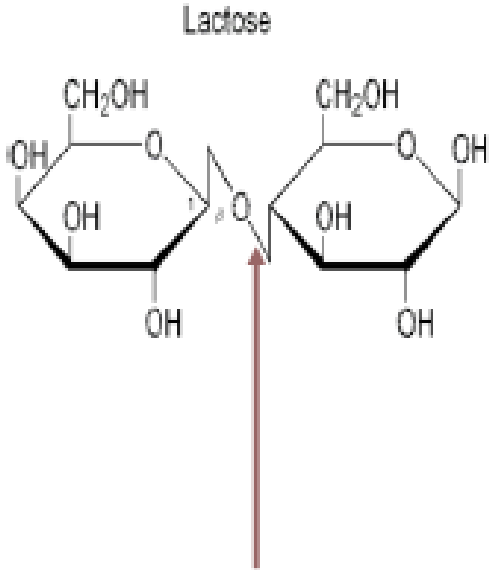
- Glycosidic bond can be N-linked or, O-linked.

N & O GLYCOSIDIC LINKAGE



Adenosine

N-Glycosidic linkage

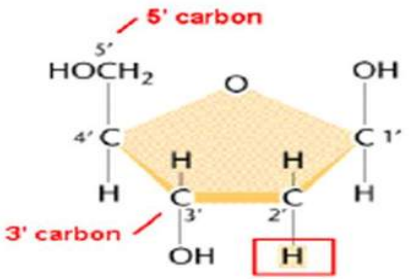


O- Glycosidic linkage

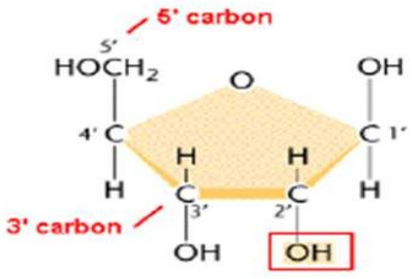
Significance of Monosaccharides

DEOXY SUGARS

- Oxygen of the hydroxyl group is removed to form deoxy sugars.
- Non reducing and non osazone forming.
- Important part of nucleic acids.



2-Deoxyribose



Ribose

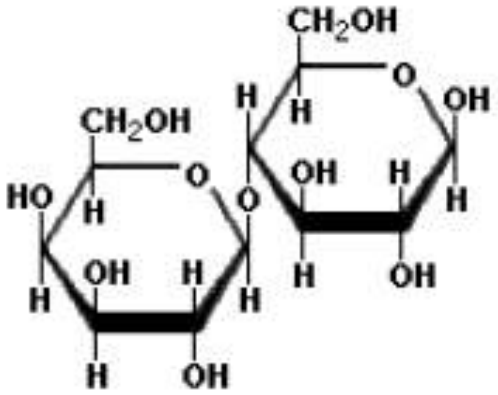
DISACCHARIDES

- Two monosaccharides combined together by glycosidic linkage.
  - Reducing : Maltose, Lactose – with free aldehyde or keto group.
  - Non-reducing : Sucrose, Trehalose – no free aldehyde or keto group.

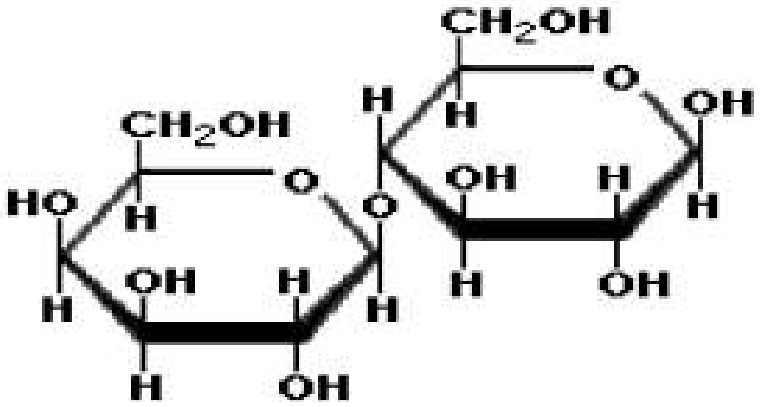
SUCROSE

- Cane sugar.
- α-D-glucose & β-D-fructose units held together by (α1→ β2) glycosidic bond.
- Reducing groups in both are involved in bond formation, hence non reducing.





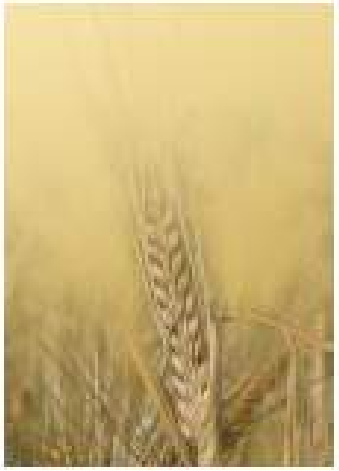
LACTOSE



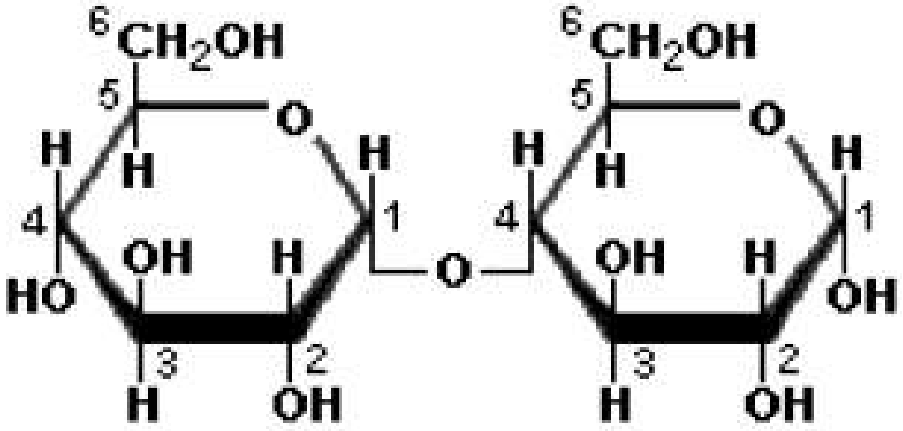


- Present in milk.
- β-D-galactose & β-D-glucose units held together by β (1→4) glycosidic bond.

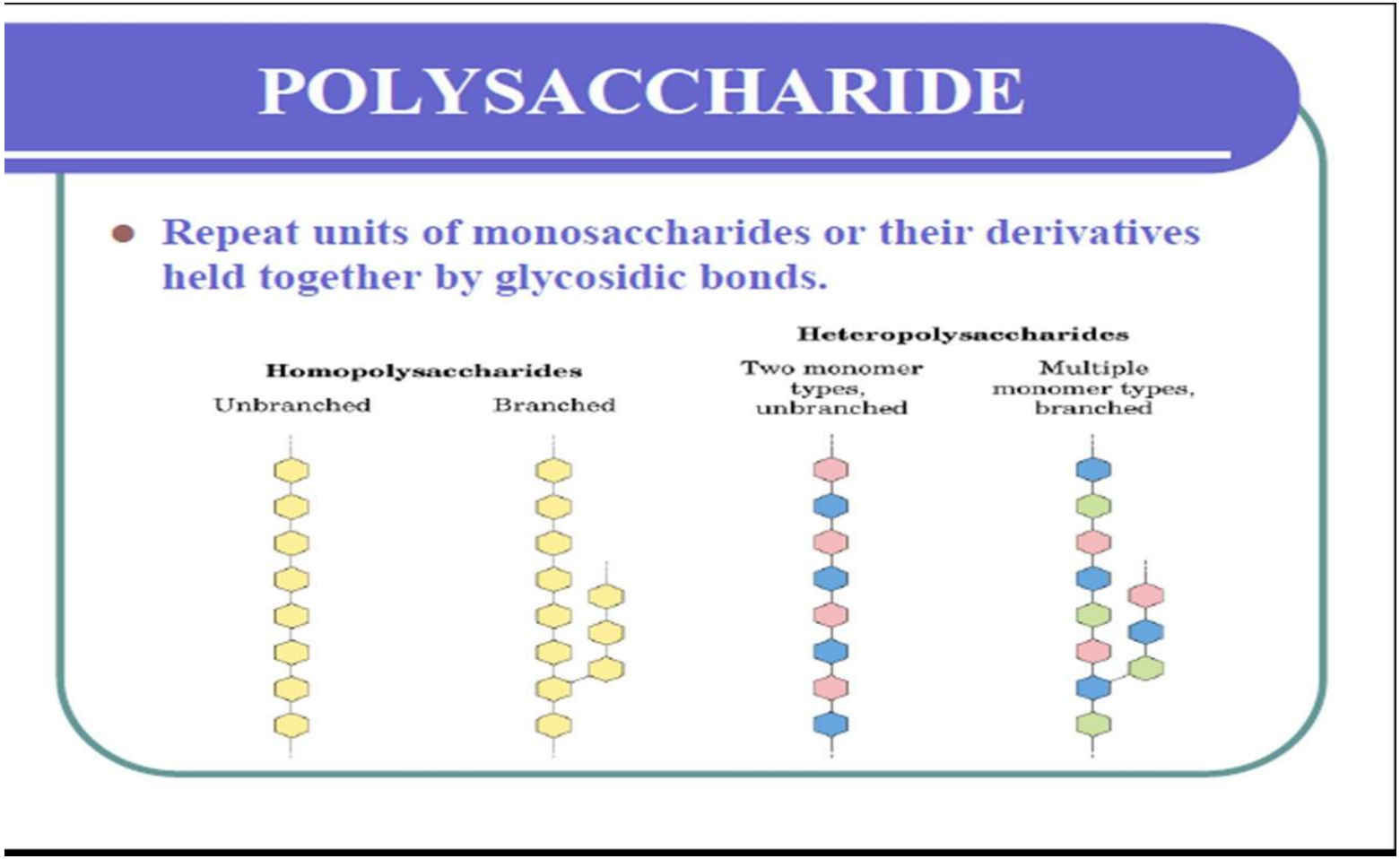
MALTOSE



- Malt sugar.
  Produced during the course of digestion of starch by the enzyme amylase.
- Two α-D-glucose units held together by α (1→4) glycosidic bond.

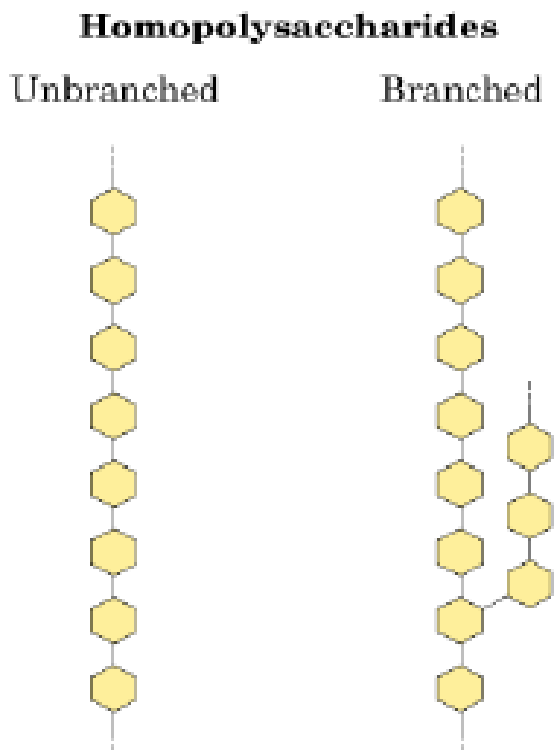


Significance of polysaccharides

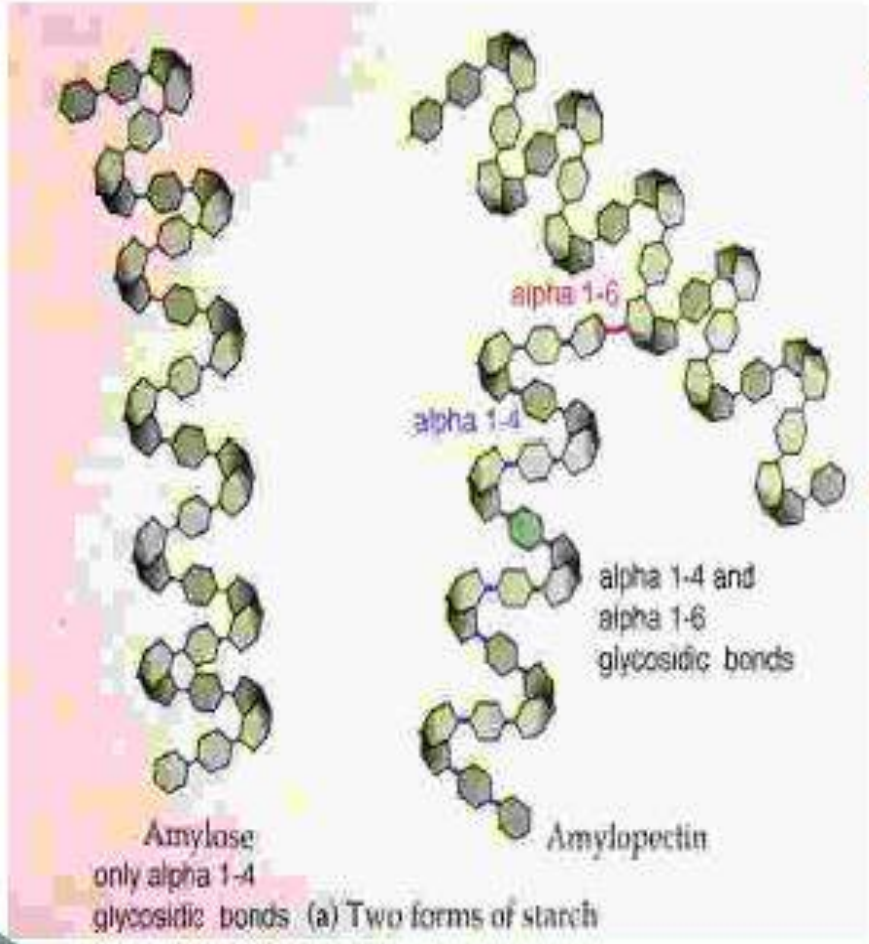


HOMOPOLYSACCHARIDES

- Starch
- Glycogen
- Cellulose
- Inulin
- Dextrans
- Chitin



STARCH

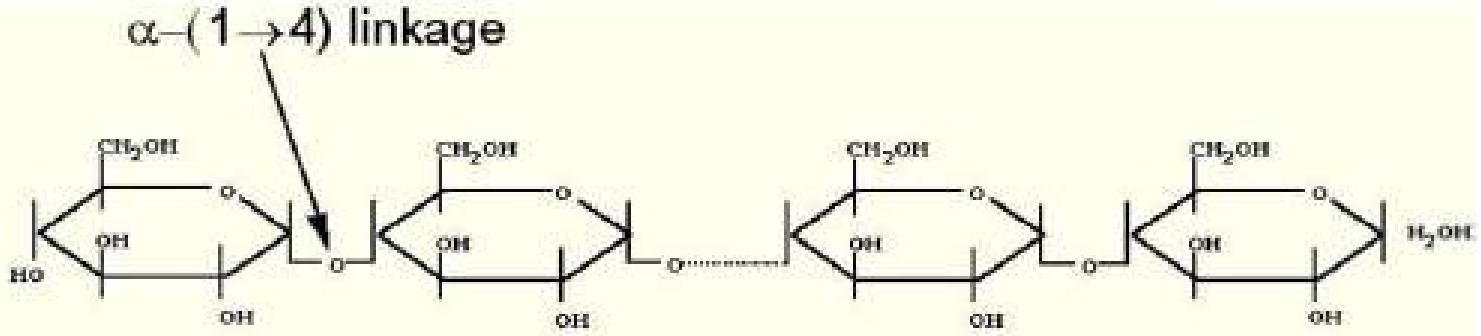


(a) Two forms of starch

- Carbohydrate reserve of plants. Present in Cereals, Roots, Tuber, Vegetables.
- Consists of Amylose (water soluble) & Amylopectin (water insoluble).

AMYLOSE

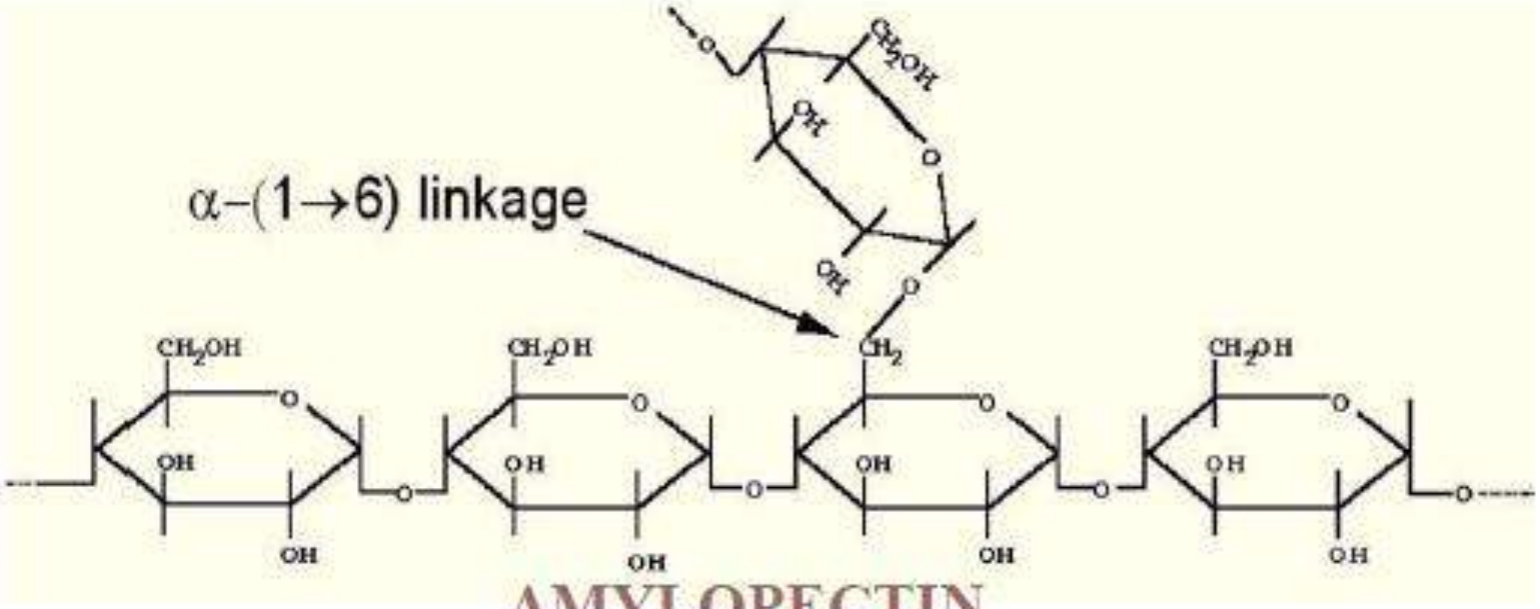
- Long unbranched chain.
- 200 – 20,000 D-glucose units held together by α (1→4) glycosidic linkages.



AMYLOSE

AMYLOPECTIN

- Branched chain. (α 1→6 glycosidic bonds at branches).
- 20 – 30 glucose units per branch.



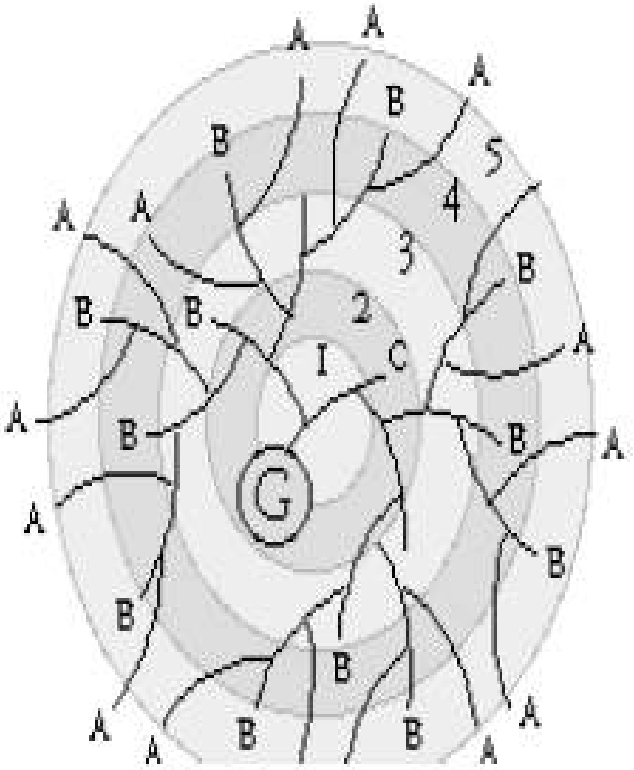
AMYLOPECTIN

ACTION OF AMYLASE

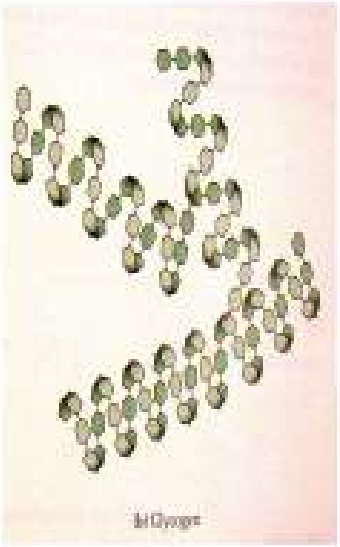
- Starch —(Salivary & pancreatic α-amylase or ß-amylase)→ Dextrins → α/ß-Maltose
- Amylopectin —(ß-amylase)→ Limit dextrin + Maltoses

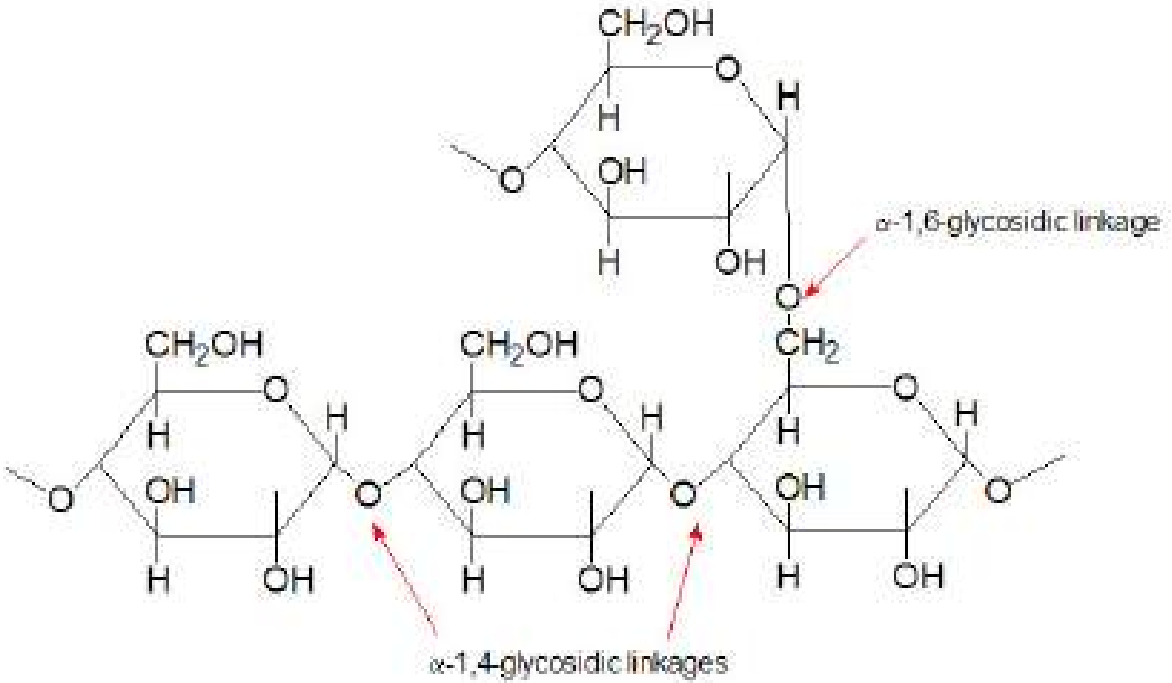
GLYCOGEN

- Reserve carbohydrate in animals. Stored in liver & muscle.
- Forms red-brown/brown-violet colour with iodine.
- Contains primer protein: Glycogenin.
- More branched and compact than amylopectin. Every 11th sugar molecule has a branch.

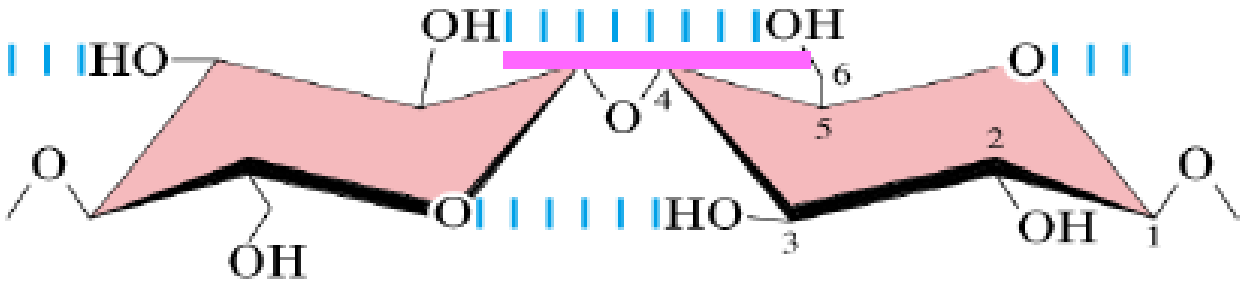


GLYCOGEN STRUCTURE





CELLULOSE

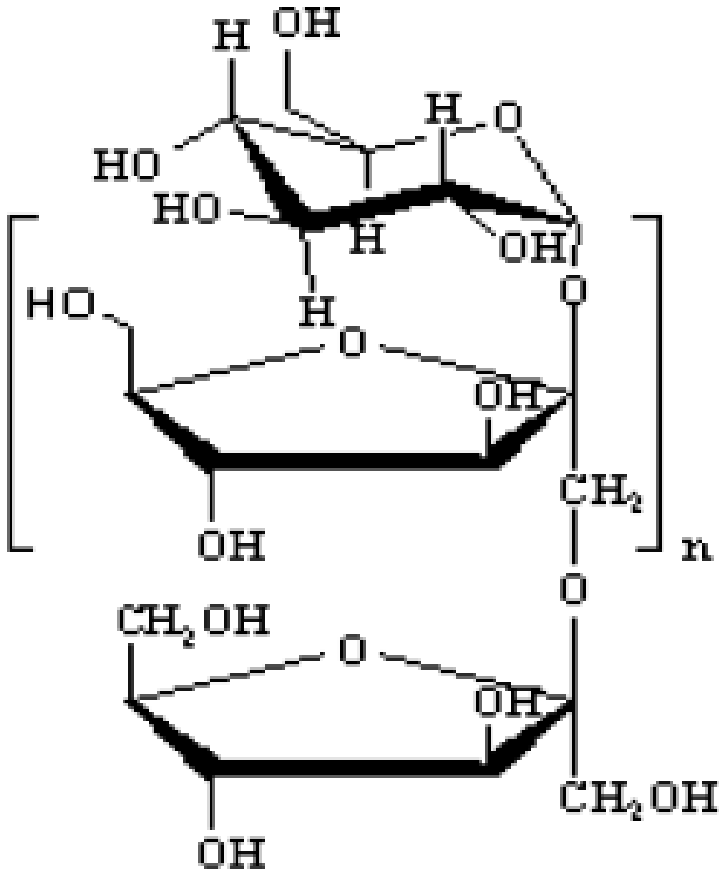


(β1→4)-linked D-glucose units

(a)

- Chief carbohydrate in plants.
- Made up of glucose units combined with cellobiose bridges.
- No branching point.
- Cannot be digested by human due to absence of Cellobiase.

INULIN

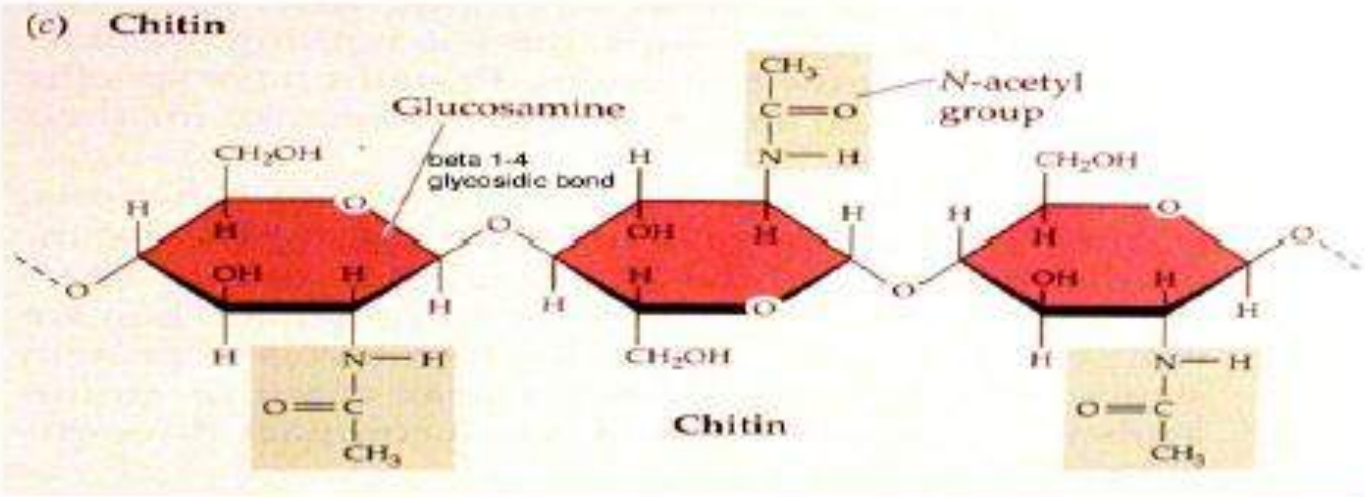


- Inulin is made up of D-fructose units with repeating ß-1,2 linkages.
- It acts as a marker for glomerular filtration since it is not synthesized, metabolized but filtered completely by glomerulus.

CHITIN

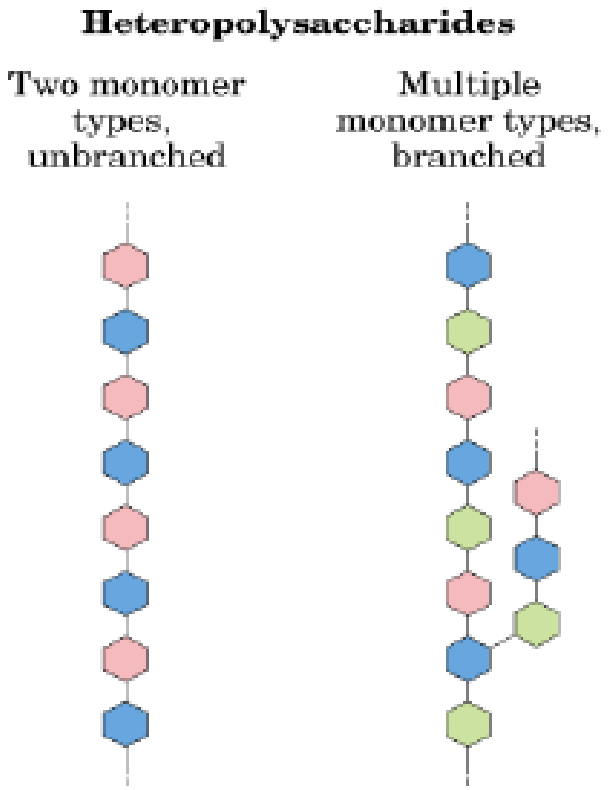
- Chitin is found in crustaceans eg.lobsters,crabs,shrimps,insects.
- Composed of N-acetyl glucosamine units joined by ß-1,4 glycosidic linkages.





HETEROPOLYSACCHARIDES

- **Agar**
- **Mucopolysaccharides:**
  - ✓ **Hyaluronic acid**
  - ✓ **Heparin**
  - ✓ **Chondroitin sulphate**
  - ✓ **Keratan sulphate**
  - ✓ **Dermatan sulphate**



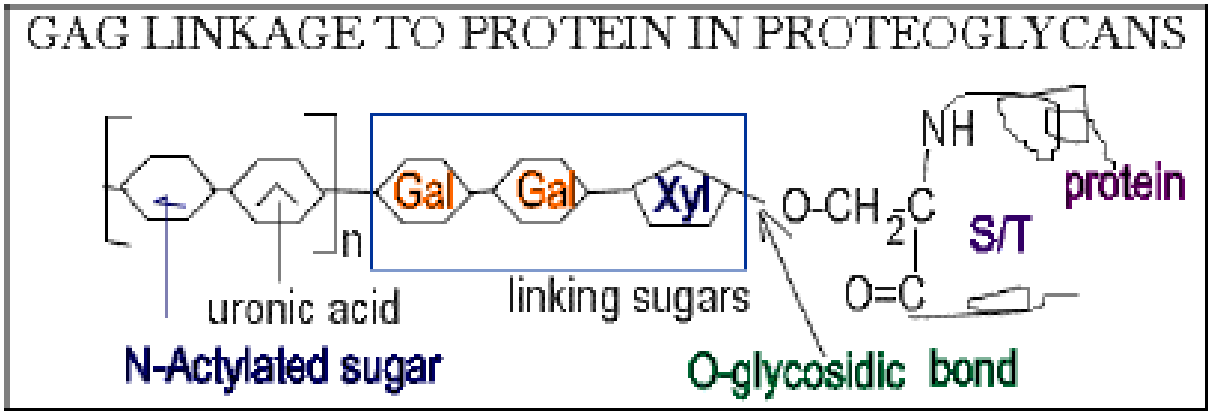
AGAR



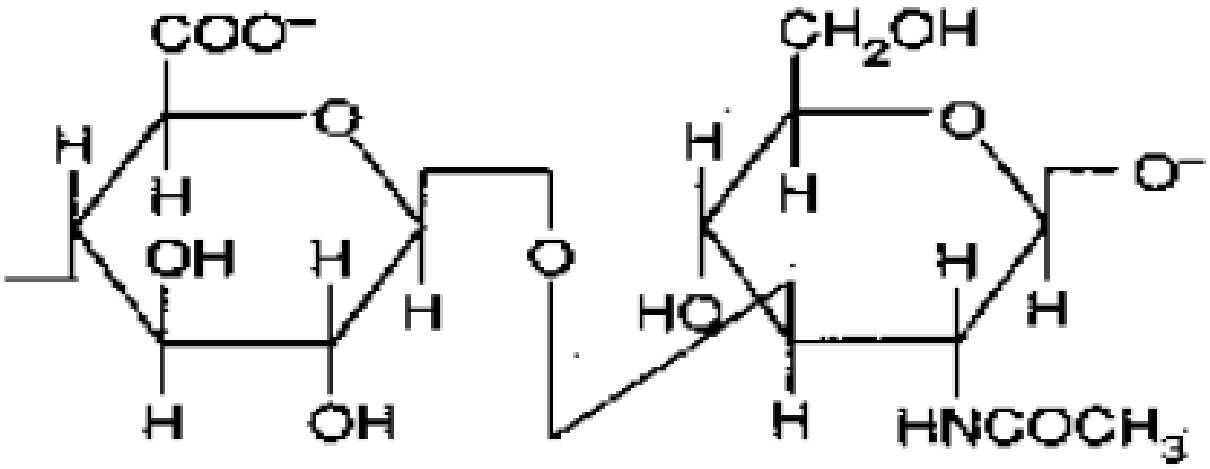
- Prepared from sea weeds.
- Contains Galactose, Glucose and other sugars.
- Used as supporting medium for immunodiffusion & immunoelectrophoresis.
- Agarose contains Galactose combined with3,6 anhydrogalactose units.
- Agarose is used as matrix for electrophoresis.

MUCOPOLYSACCHARIDES

- Also known as GAG.
- Made up of repeating units of sugar derivatives (aminosugars and uronic acids).
- Acetylated amino groups, sulfates and carboxyl groups are generally present.

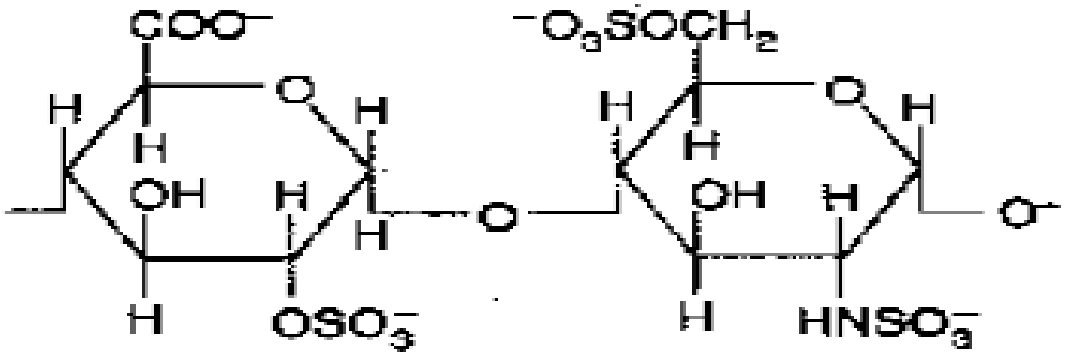


HYALURONIC ACID



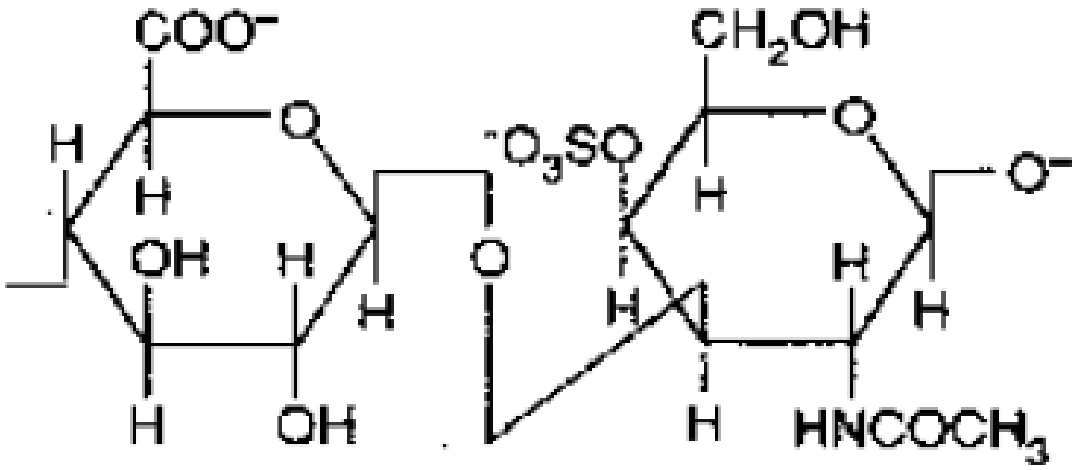
- Present in connective tissues, tendons, synovial fluid and vitreous humor.
- Composed of repeating units of N-acetyl glucosamine → ß-1,4 glucuronic acid → ß-1,3 N-acetyl glucosamine.

HEPARIN



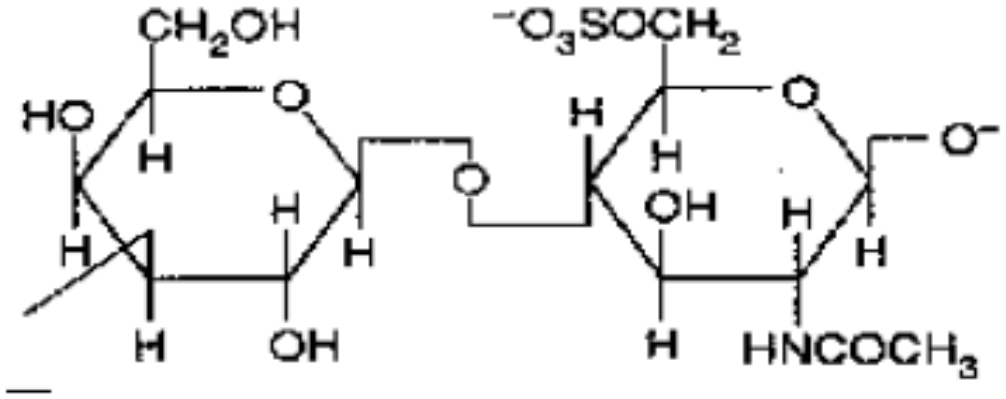
- Anticoagulant. Bind and activate Antithrombin III, which in turn activates Thrombin, Factor X & Factor IX.
- Present in lung, spleen and monocytes.
- Contains repeating units of sulphated glucosamine → α-1,4 L-iduronic acid.
- Sulphated: Heparan sulphate

CHONDROITIN SULPHATE



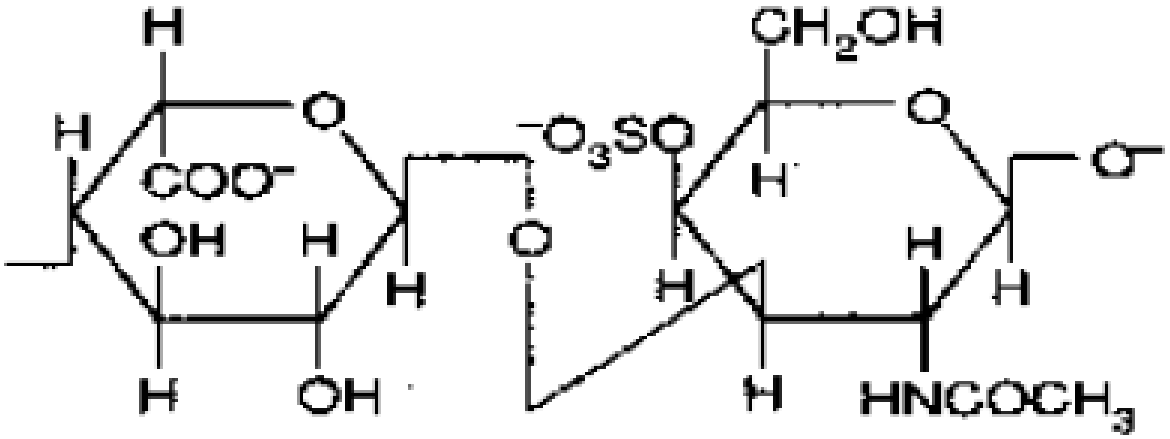
- Present in ground substances of connective tissues of cartilages, bones & tendons.
- Composed of Glucuronic acid → ß-1,3 N-acetyl galactosamine sulphate → ß-1,4 and so on.

KERATAN SULPHATE



- Only GAG not having Uronic acid.
- Found in cornea and tendons.
- Repeating units are Galactose & N-acetyl galactosamine in ß linkage.

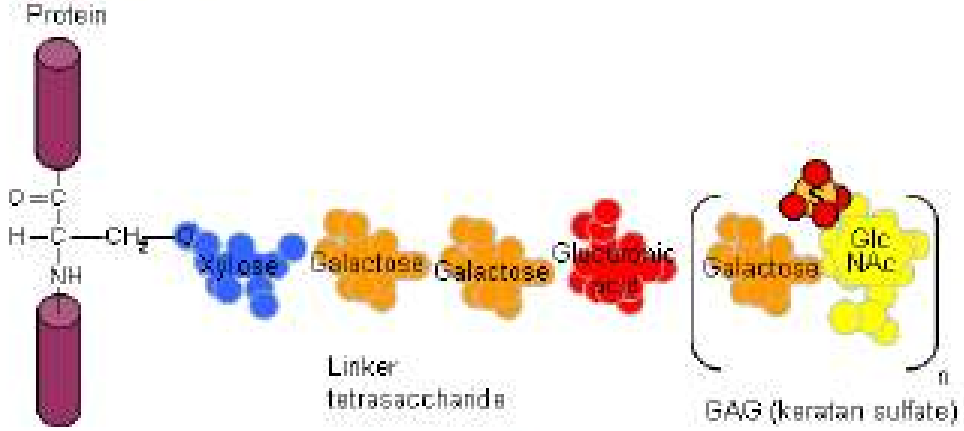
DERMATAN SULPHATE



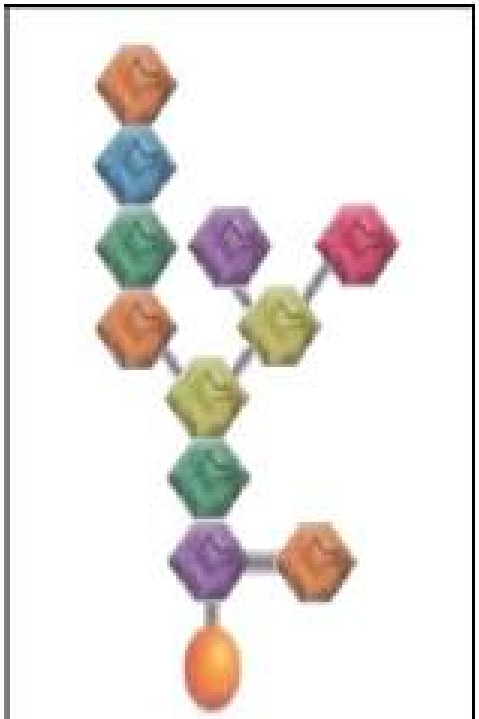
- Found in skin, blood vessels & heart vessels.
- Contains L-iduronic acidand N-acetyl galactosamine in ß-1,3 linkage.

PROTEOGLYCANS & GLYCOPROTEINS

- **Proteoglycans:** When carbohydrate chains are attached to a polypeptide chain.
- **Glycoproteins:** Carbohydrate content ≤ 10%.
- **Mucoprotein:** Carbohydrate content ≥10%



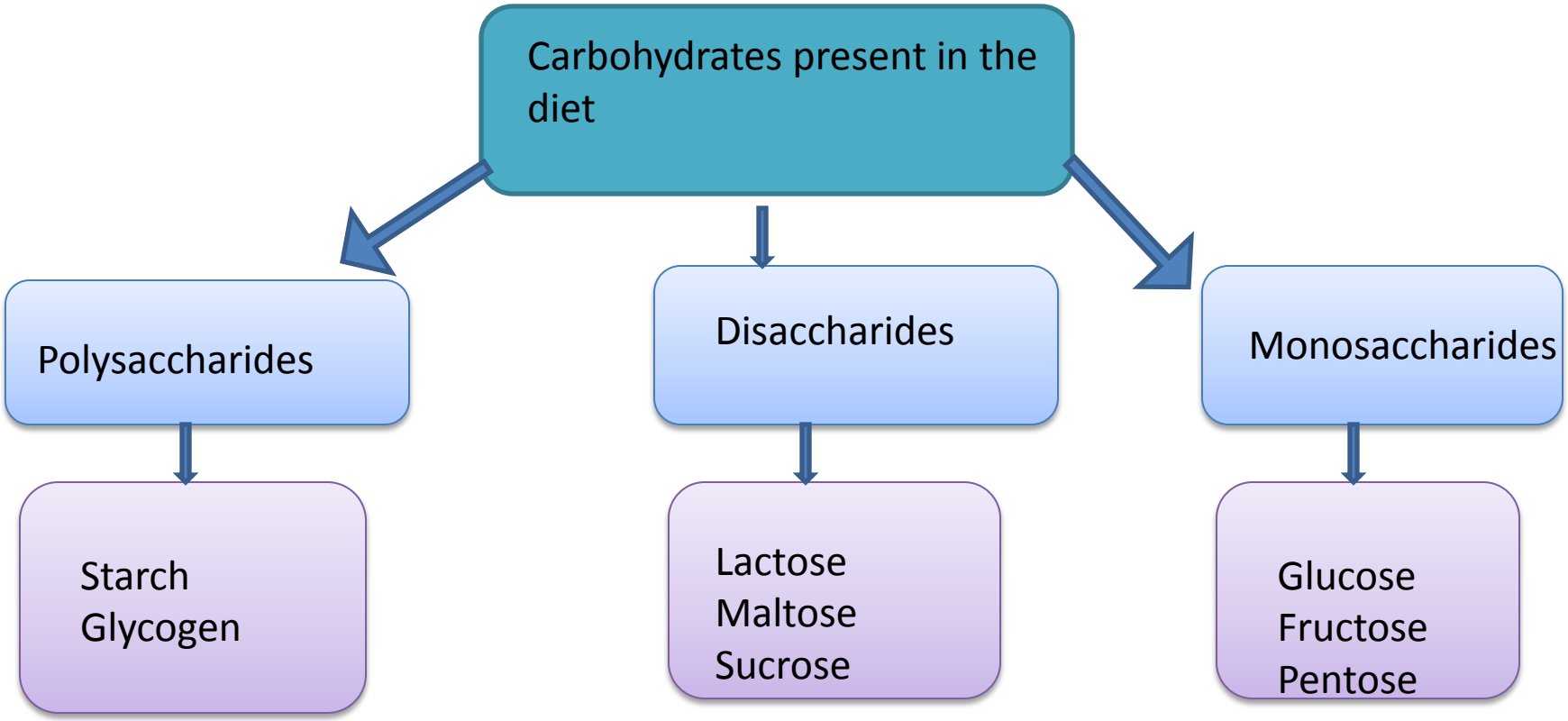
Proteoglycan



Glycoform attached to a protein (Glycoprotein)

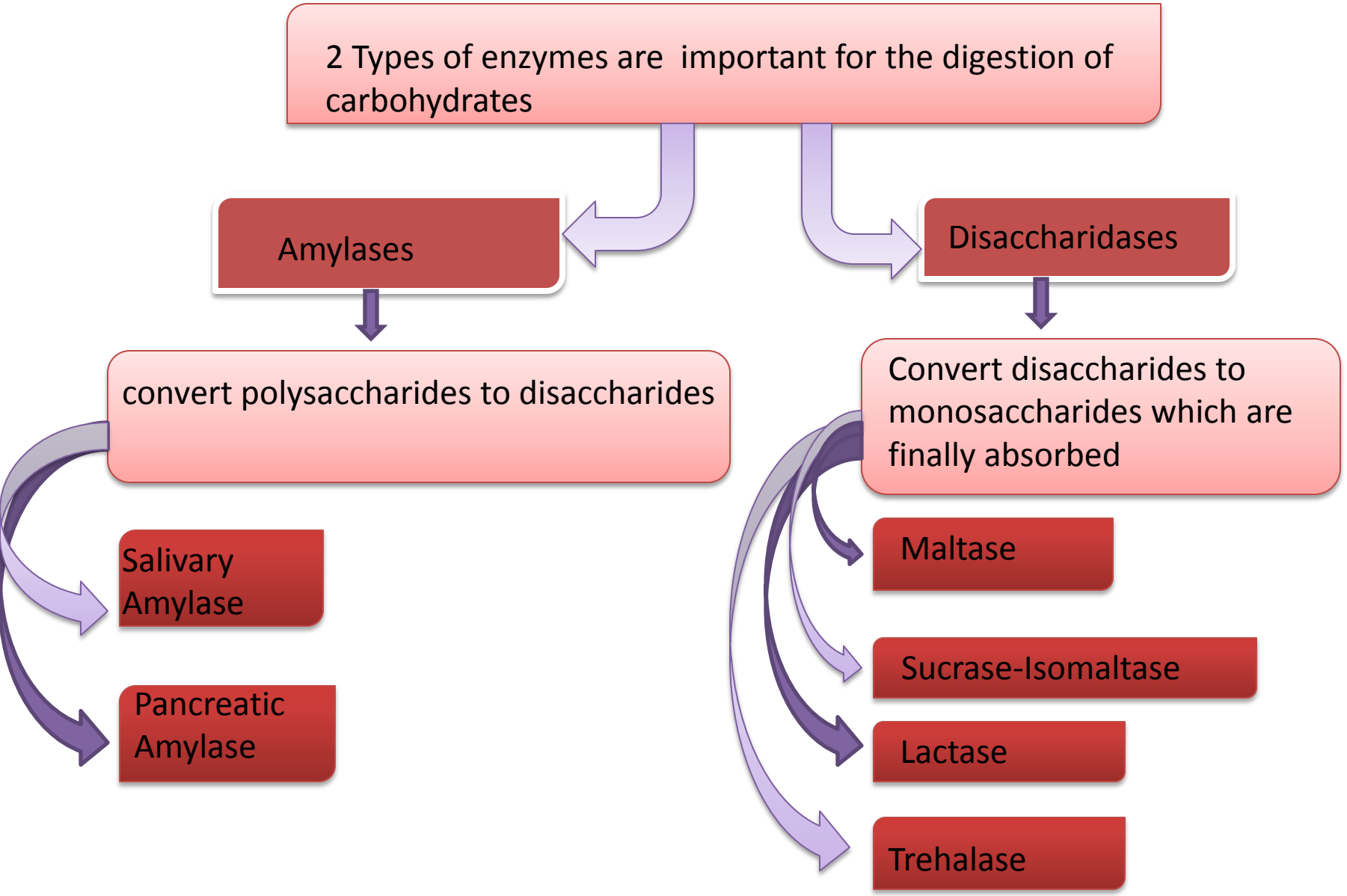
Digestion and Absorption of carbohydrates with *clinical significance*.

Digestion and absorption of carbohydrates

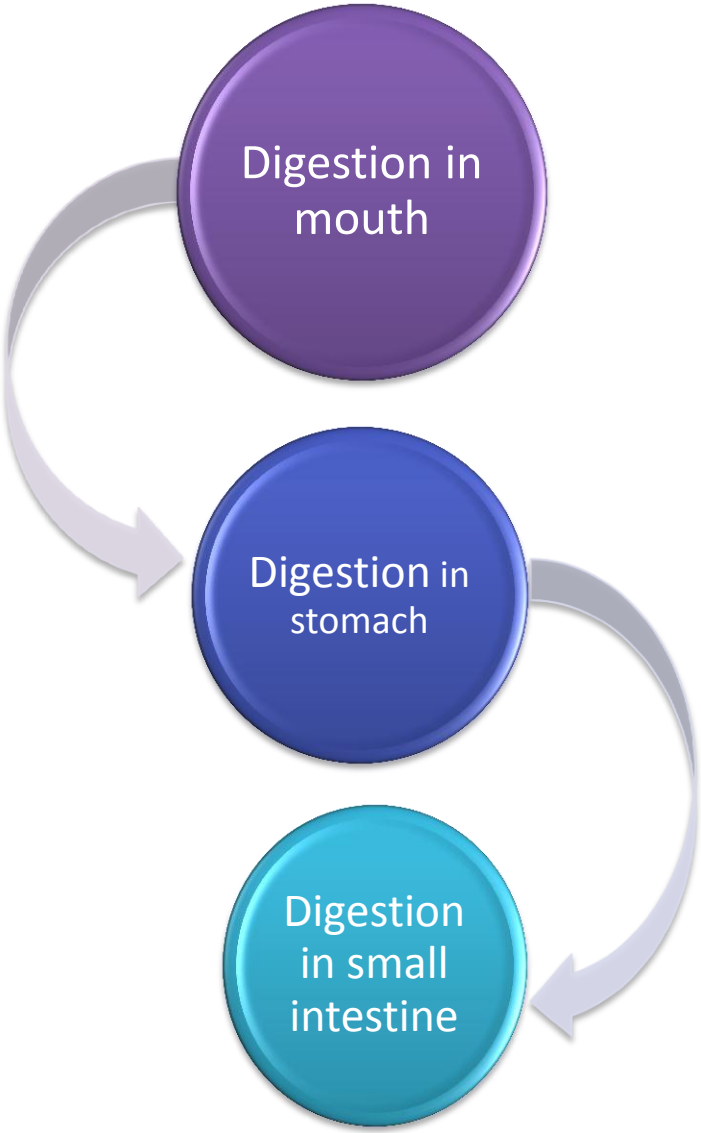


In GIT, all complex carbohydrates are converted to simpler monosaccharide form which is the absorbable form.

Details of digestion of carbohydrates



DIGESTION OF CARBOHYDRATES



Digestion in the Mouth

- Digestion of Carbohydrate starts in the mouth, upon contact with saliva during mastication.

Saliva contains a carbohydrate splitting enzyme called ***salivary amylase ,*** also known as ***ptylin***.

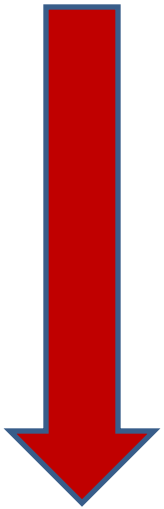


Action of ptylin (salivary amylase)

- Location: mouth
- It is α-amylase and requires Cl^- ion for activation with an optimum pH of 6.7 (Range 6.6 to 6.8).
- The enzyme hydrolyses α-1→ 4 glycosidic linkages deep inside polysaccharide molecules.
- However, ptylin action stops in the stomach when the pH falls to 3.0.

Starch, Glycogen and dextrins

(Large polysaccharide molecules)



α- Amylase

Glucose,Maltose and Maltotriose.

(Smaller molecules)



Drawback

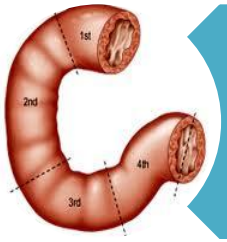
- **Shorter duration of food in mouth.**
- **Thus it is incomplete digestion of starch or glycogen in the mouth**



Digestion in the Stomach

- **There is no enzyme to break the glycosidic bonds in gastric juice.**
- **However, HCl present in the stomach causes hydrolysis of sucrose to fructose and glucose.**

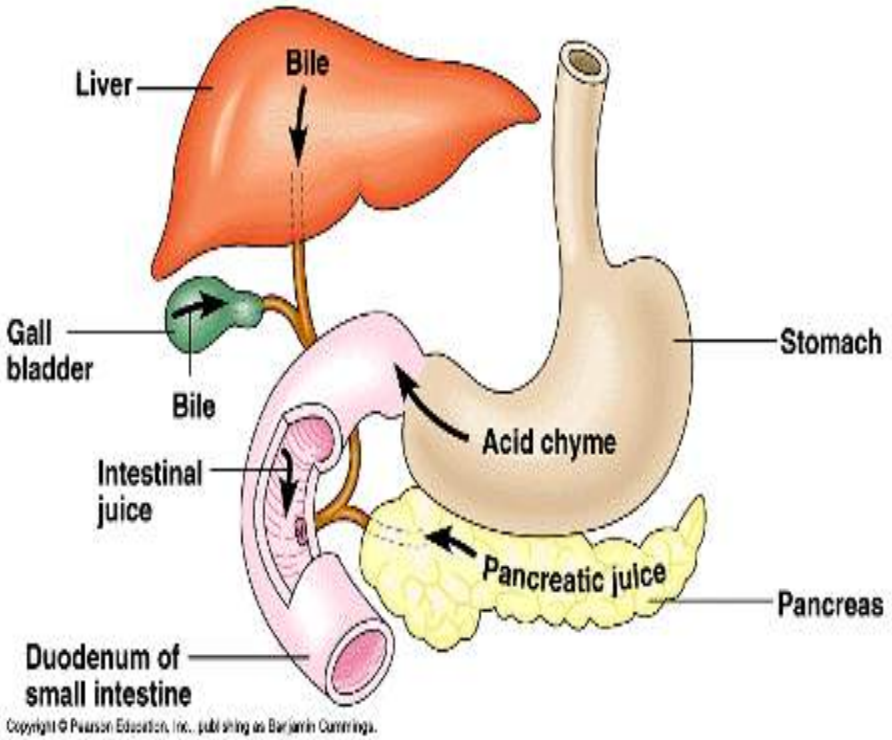
Sucrose $\xrightarrow{HCl}$ **Fructose + Glucose**



Digestion in Duodenum

- Food bolus reaches the duodenum from the stomach where it meets the pancreatic juice.
- Pancreatic juice contains a carbohydrate splitting enzyme,

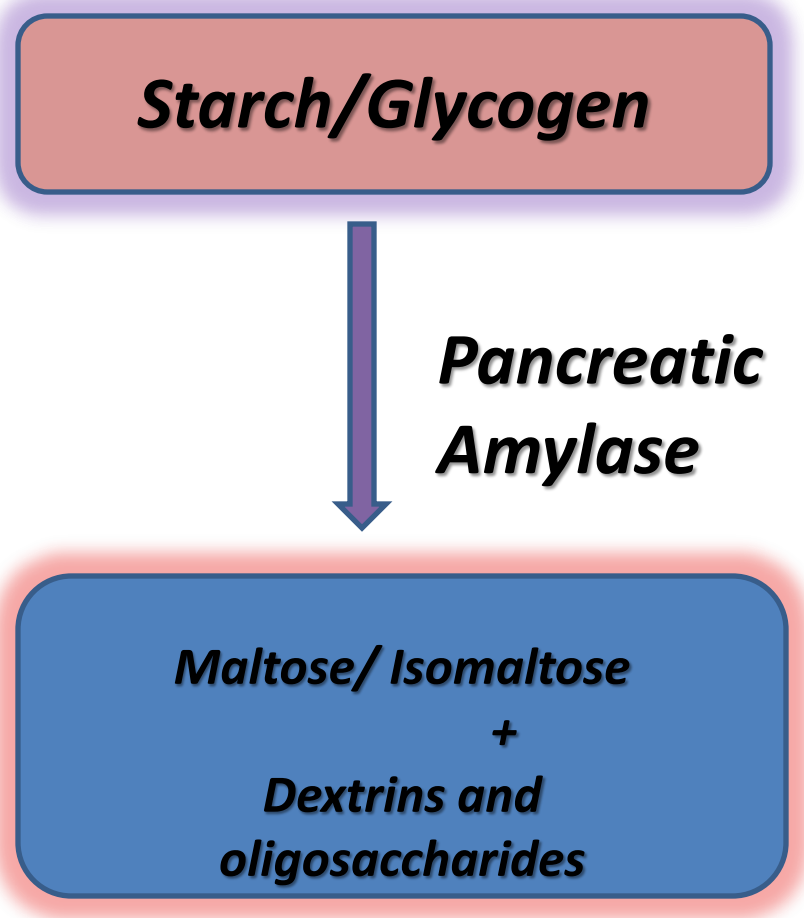
pancreatic amylase **(amylopsin)** similar to salivary amylase.



Action of pancreatic amylase

- **It is an *α- Amylase***
- ***Optimum pH=7.1***
- ***Like ptylin, it requires*** **Cl^- ion for its activity.**
- **It hydrolyses α-1→ 4 glycosidic linkages situated well inside polysaccharide molecules.**
- **Note:** *Pancreatic amylase, an isoenzyme of salivary amylase, differs only in the optimum pH of action. Both the enzymes require Chloride ions for their actions (Ion activated enzymes).*

Reaction catalyzed by pancreatic amylase

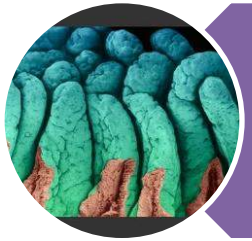




Digestion in Small Intestine

Note:

- **Main digestion takes place in the small intestine by pancreatic amylase**
- **Digestion is completed by pancreatic amylase because food stays for a longer time in the intestine.**



What are Disaccharidases?

- They are present in the brush border epithelium of intestinal mucosal cells where the resultant monosaccharides and others arising from the diet are absorbed.

- The different disaccharidases are :

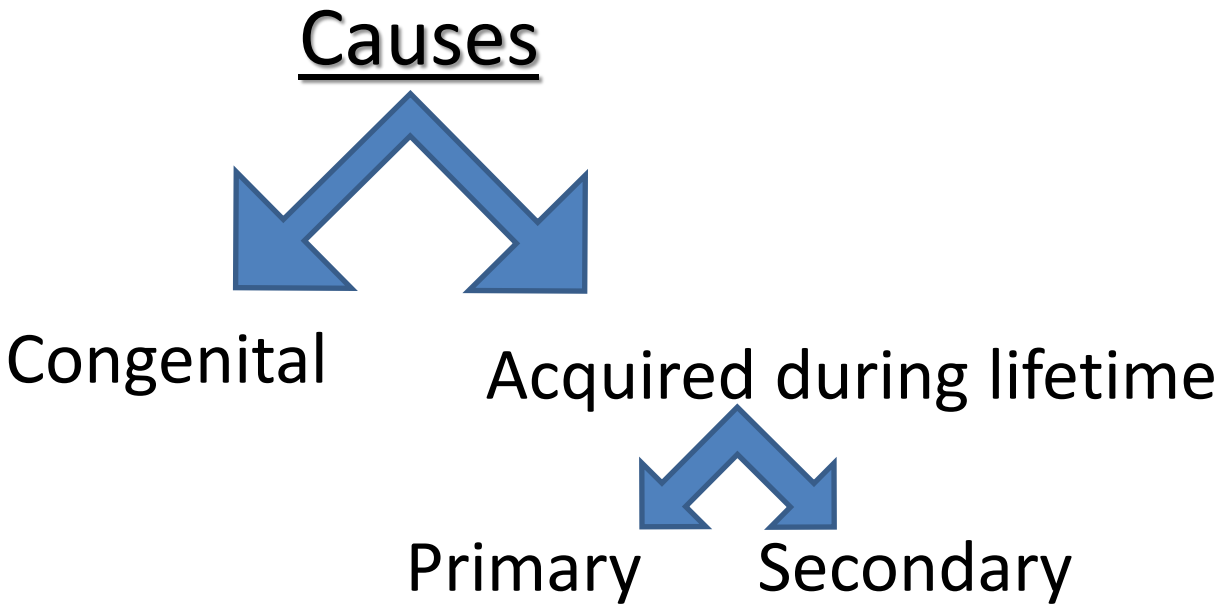
1) Maltase,
2) Sucrase-Isomaltase (a bifunctional enzyme catalyzing hydrolysis of sucrose and isomaltose)
3) Lactase

Reactions catalyzed by Disaccharidases

- Maltose —(Maltase)→ Glucose + Glucose
- Sucrose Isomaltose —(Sucrase Isomaltase)→ 3Glucose + fructose
- Lactose —(Lactase)→ Glucose + Galactose

Clinical significance of Digestion

- Lactose intolerance is the inability to digest lactose due to the deficiency of Lactase enzyme.
-





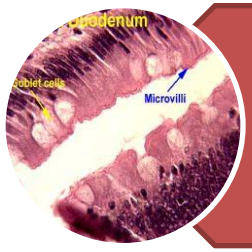
Congenital Lactose intolerance

- **It is a congenital disorder**
- **There is complete absence or deficiency of lactase enzyme.**
- **The child develops intolerance to lactose immediately after birth.**
- **It is diagnosed in early infancy.**
- **Milk feed precipitates symptoms.**



Primary Lactase deficiency

- Primary lactase deficiency develops over time
- There is no congenital absence of lactase but the deficiency is precipitated during adulthood.
- The gene for lactose is normally expressed upto RNA level but it is not translated to form enzyme.
- It is very common in Asian population.
- There is intolerance to milk + dairy products.



Secondary lactase deficiency

- It may develop in a person with a healthy small intestine during episodes of acute illness.
- This occurs because of mucosal damage or from medications resulting from certain **gastrointestinal diseases**, including exposure to **intestinal parasites** such as Giardia lamblia.
- In such cases the production of lactase may be permanently disrupted.
- A very common cause of **temporary** lactose intolerance is gastroenteritis, particularly when the gastroenteritis is caused by **rotavirus.**
- Another form of temporary lactose intolerance is lactose overload in infants. Secondary lactase deficiency also results from injury to the small intestine that occurs with **celiac disease, Crohn's disease, or chemotherapy**.
- This type of lactase deficiency can occur at any age but is more common in infancy.



Clinical manifestations

- In the form of abdominal cramps, distensions, diarrhea, constipation, flatulence upon ingestion of milk or dairy products

Biochemical basis

- Undigested lactose in intestinal lumen is acted upon by bacteria and is converted to CO_2 , H_2 , 2 carbon compounds and 3 carbon compounds or it may remain undigested.



- CO_2 and H_2 causes Distensions and flatulence
- Lactose + 2C + 3C are osmotically active.

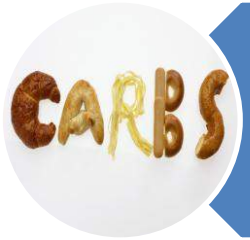
- They withdraw H_2O from intestinal mucosal cell and cause osmotic diarrhea or constipation because of undigested bulk.



Abdominal distension



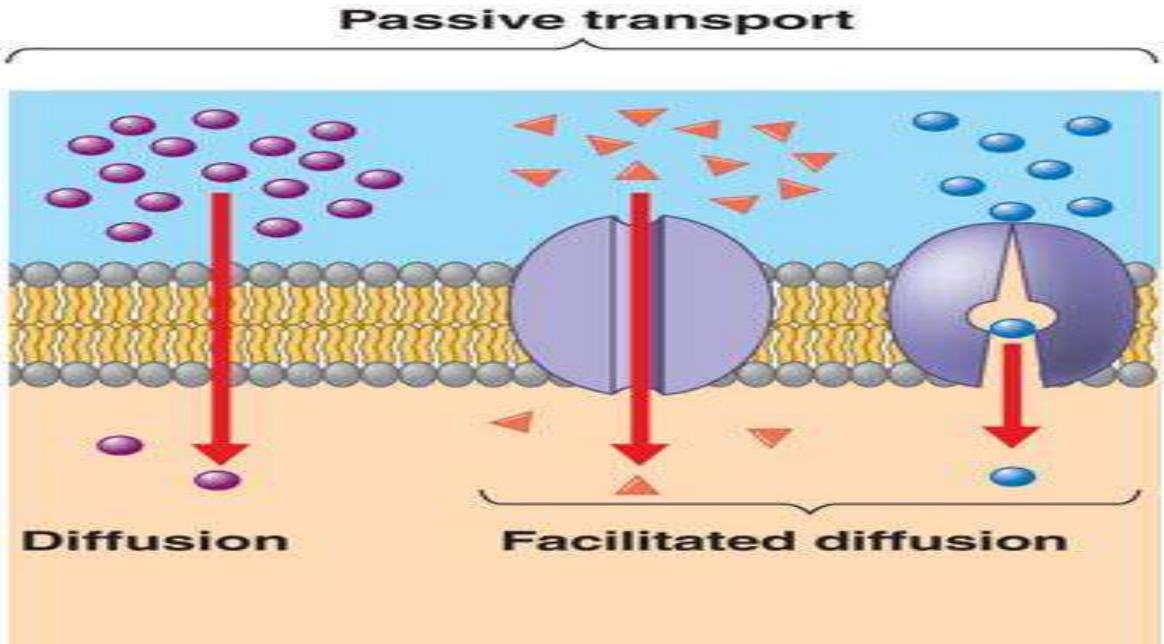
Flatulence

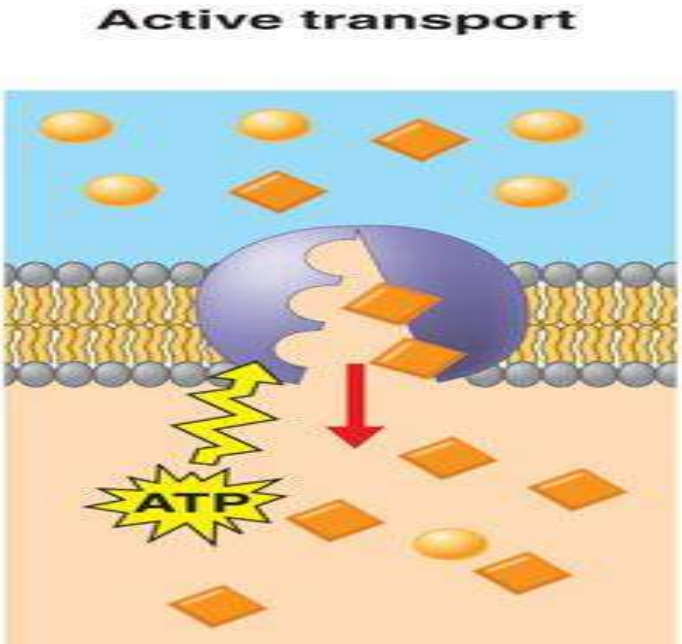


Absorption of carbohydrates

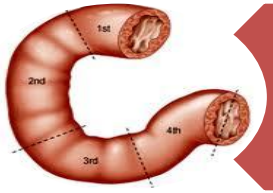
3 mechanisms

- Passive diffusion
- Facilitated diffusion/Carrier mediated
- Active transport





Features	Passive diffusion	Facilitated diffusion	Active transport
Concentration gradient	Down the concentration gradient from high to low.	Down the concentration gradient from high to low.	Against a concentration gradient from low to high
Energy expenditure	none	none	Energy expenditure is in the form of ATP
Carrier protein/ transporter	Not required	required	required
Speed	Slowest mode	Fast	Fastest mode



Glucose transporters

Glucose transporters

2 types

Na^+ dependent transporter

Also called

SGLT

Na^+ independent transporter

Also called

GLUT

Thank you